

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=1.412937660205671e-07

initial={'cost': 1.4206294312964576, 'gradient_inf_norm': 0.0377105352581939, 'gradient_l2_norm': 0.05415474848763546, 'jacobian': {'condition_number_nonzero_subspace': 405.3061925815292, 'numerical_rank': 14, 'numerical_tolerance': 9.540767286417267e-11, 'singular_values': [119.28871736795509, 118.13316616931783, 102.64851500440028, 48.60611998465352, 44.72601485169613, 25.867565316880995, 12.98883716392602, 11.72055696278085, 11.324756502423742, 11.045939607104708, 9.878998819233527, 3.7982312942476866, 3.7927787000184057, 0.2943175296882729]}, 'residual_l2_norm': 1.685603412013904}

final={'cost': 1.420628905726001, 'gradient_inf_norm': 1.412937660205671e-07, 'gradient_l2_norm': 2.0921343218342325e-07, 'jacobian': {'condition_number_nonzero_subspace': 405.0893028901442, 'numerical_rank': 14, 'numerical_tolerance': 9.540695944606147e-11, 'singular_values': [119.28782537752205, 118.13186225386, 102.648330362545, 48.60621398727305, 44.725990220826276, 25.867545293755164, 12.988875396783898, 11.720279331439366, 11.32446007017322, 11.045614258443992, 9.87871202958631, 3.7982219245629794, 3.79278574738097, 0.29447290887824706]}, 'residual_l2_norm': 1.685603100214283}

coordinate_step_l2_norm=0.00028345052051270204

cost_reduction=5.255704564888219e-07

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=7.33331540232405e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 87.31736356979816, 'difference_l2_norm': 6.403257671567297e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 87.31736356965725, 'relative_error': 7.33331540232405e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.